

**Multiple Choice:**

1. The maximum synchronizing capability in coding techniques is present in:
A. Manchester format
B. Polar NRZ
C. Polar RZ
D. Polar quaternary NRZ
2. What is entropy?
A. Average information per message
B. Information in a signal
C. Amplitude of signal
D. All of the choices
3. For decoding in convolution coding, in a code tree,
A. Diverge upward when a bit is 0 and diverge downward when the bit is 1
B. Diverge downward when a bit is 0 and diverge upward when the bit is 1
C. Diverge left when a bit is 0 and diverge right when the bit is 1
D. Diverge right when a bit is 0 and diverge left when the bit is 1
4. Run length encoding is used for
A. Reducing the repeated string of characters
B. Bit error correction
C. Correction of error in multiple bits
D. All of the choices
5. For M equally likely messages, $M \gg 1$, if the rate of information $R \leq C$, the probability of error is _____.
A. arbitrarily small
B. close to unity
C. not predictable
D. unknown
6. The characteristics of compressor in μ -law companding are:
A. Continuous in nature
B. Logarithmic in nature
C. Linear in nature
D. Discrete in nature
7. The digital modulation technique in which the step size is varied according to the variation in the slope of the input is called _____.
A. Delta modulation
B. PCM
C. Adaptive delta modulation *
D. PAM
8. BPSK system modulates at the rate of _____ bit/symbol.
A. 1 *
B. 2
C. 3
D. 4
9. In uniform quantization process
A. The step size remains same
B. Step size varies according to the values of the input signal
C. The quantizer has linear characteristics
D. Both the step size remains same and the quantizer has linear characteristics are correct *
10. In polar RZ format for coding, symbol '0' is represented by:
A. Zero voltage
B. Negative voltage
C. Pulse is transmitted for half the duration
D. Both negative voltage and pulse is transmitted for half the duration are correct
11. Equalization in digital communication:
A. Reduces inter symbol interference
B. Removes distortion caused due to channel
C. Is done using linear filters
D. All of the choices
12. QPSK is a modulation scheme where each symbol consists of _____ bits
A. 4
B. 2
C. 1
D. M number of bits, depending upon the requirements
13. DPCM is a technique:
A. To convert analog signal into digital signal
B. Where difference between successive samples of the analog signals are encoded into n-bit data streams
C. Where digital codes are the quantized values of the predicted value
D. All of the choices
14. In DPSK technique, what is the technique used to encode bits?
A. AMI
B. Differential code *
C. Uni polar RZ format
D. Manchester format
15. The criterion used for pulse shaping to avoid ISI is _____.
A. Nyquist criterion *
B. Quantization
C. Sample and hold
D. PLL
16. The code in convolution coding is generated using:
A. EX-OR logic
B. AND logic
C. OR logic
D. None of the choices
17. Minimum shift keying is similar to:
A. Continuous phase frequency shift keying
B. Binary phase shift keying
C. Binary frequency shift keying
D. QPSK
18. The polarities in NRZ format use:
A. Complete pulse duration
B. Half duration
C. Both positive as well as negative value
D. Each pulse is used for twice the duration



19. In coherent detection of signals,

- A. Local carrier is generated
- B. Carrier of frequency and phase as same as transmitted carrier is generated
- C. The carrier is in synchronization with modulated carrier
- D. All of the choices

END

20. The maximum data transmission rate in T1 carrier system is _____ Mbps.

- A. 2.6
- B. 1000
- C. 1.544
- D. 5.6

21. The capacity of Gaussian channel is

- A. $C = 2B(1+S/N)$ bits/s
- B. $C = B^2(1+S/N)$ bits/s
- C. $C = B(1+S/N)$ bits/s *
- D. $C = B(1+S/N)^2$ bits/s

22. The sequence of operations in which PCM is done:

- A. Sampling, quantizing, encoding
- B. Quantizing, encoding, sampling
- C. Quantizing, sampling, encoding
- D. None of the choices

23. Granular noise occurs when:

- A. Step size is too small
- B. Step size is too large *
- C. There is interference from the adjacent channel
- D. Bandwidth is too large

24. What is one of the disadvantages of PCM?

- A. It requires large bandwidth *
- B. Very high noise
- C. Cannot be decoded easily
- D. All of the choices

25. The process of converting the analog sample into discrete form is called _____.

- A. modulation
- B. multiplexing
- C. quantization
- D. sampling